Appendix for ”Generalization Analysis for Adversarial Vision Transformers”

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A NOTATION

Let $[L] = \{1, 2, \ldots, L\}$. Let vectors be denoted by bold lowercase letters (e.g.,$\mathbf{w}$), where individual components are written as italicized subscripts $(w_1, \ldots, w_d)$ for $\mathbf{w} \in \mathbb{R}^d$. Matrices are designated by bold uppercase letters (e.g., $\mathbf{W}$). The (p, q)-norm of matrix $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_{d2}]$ is defined as: $\left\|\left(\|\mathbf{w}_1\|_p, \ldots, \|\mathbf{w}_{d2}\|_p\right)\right\|_q$, where $\|\cdot\|_p$ denotes the standard ℓ_p-norm. Additionally, we denotes the i column of $\mathbf{W}$ as $\mathbf{W}_{*i}$. For ease of exposition, we summarize the notations in Table 1.

Table 1: Summary of notations involved in this paper.

Notations	Meaning
$[L]$	The set of positive integers, i.e., $[L] = \{1, \ldots, L\}$.
$\mathbf{w}$	A vector in $\mathbb{R}^d$, with components $w_1, \ldots, w_d$.
$\mathbf{W}_{*i}$	The i-th column of matrix $\mathbf{W}$.
y	The class label, $y \in [K]$.
$\mathbf{X}^i$	The i-th data point, containing M tokens: $\mathbf{X}^i = \{\mathbf{x}_1^i, \ldots, \mathbf{x}_M^i\} \in \mathbb{R}^{M\times d}$.
$\mathcal{D}$	The unknown data distribution.
$f(\mathbf{X}^i)$	The prediction of the model for data $\mathbf{X}^i$.
$\mathcal{B}_\infty^\epsilon(\mathbf{X})$	The ℓ_∞-perturbed neighborhood of $\mathbf{X}$, defined as $\{\widetilde{\mathbf{X}} = [\widetilde{\mathbf{x}}_1, \ldots, \widetilde{\mathbf{x}}_M] : \|\widetilde{\mathbf{x}} - \mathbf{x}\|_\infty \le \epsilon, \forall i \in [M]\}$.
ϵ	The maximum adversarial perturbation budget.
$\widetilde{\mathbf{X}}_*$	The optimal adversarial patch sequence.
$\mathfrak{R}_S(\mathcal{H})$	The Empirical Rademacher Complexity of $\mathcal{H}$ over sample $S = \{z_1, \ldots, z_n\}$.
σ	Rademacher random variables, i.e., $P(\sigma = +1) = P(\sigma = -1) = 0.5$.
σ	An activation function with $\sigma(0) = 0$.
L_σ	The Lipschitz constant of activation function.
$\mathbf{x}_{[\text{cls}]}$	The learnable [CLS] token.
$\ell_\gamma(\cdot, \cdot)$	The ramp loss with margin γ.
$\mathbf{W}^{(i)}$	The weights of the i-th layer in a multi-layer ViT: $\{\mathbf{W}_{QK}^{(i)}, \mathbf{W}_V^{(i)}, \mathbf{W}_C^{(i)}\}$.
$\mathbf{W}^{1:i}$	The set of weights from layer 1 to layer i.
$g_{\text{block}}^{(i)}$	The function representing the output of the i-th Transformer block.
Π_{norm}	The layer normalization operator projecting rows onto the unit ball.
$\mathcal{G}_{\text{block}}^{(L)}$	The class of L-layer Transformer blocks.
$\mathcal{G}_{\text{scalar}}^{(L)}$	The class of scalar output functions for L-layer ViTs.
Ψ_{l+1}, Ψ'_{l+1}	Terms capturing norm amplification through layers in multi-layer analysis.
$\|\mathbf{W}\|_{p,q}$	The (p, q)-norm of matrix $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_{d_2}]$, defined as $\|(\|\mathbf{w}_1\|_p, \ldots, \|\mathbf{w}_{d_2}\|_p)\|_q$.

Lemma 1. *Let $f(Z) = softmax(Z)$. For any $Z_1, Z_2 \in \mathbb{R}^d$,*

$$\|f(Z_1) - f(Z_2)\|_\infty \le \frac{1}{2}\|Z_1 - Z_2\|_\infty.$$

Proof. By the Mean Value Inequality for vector-valued functions, it suffices to show that the induced L_∞ norm of the Jacobian matrix $J(Z)$ is bounded by $1/2$ for all Z. The inequality states:

$$\|f(Z_1) - f(Z_2)\|_\infty \le \left(\sup_{Z\in\mathbb{R}^d} \|J(Z)\|_\infty\right) \|Z_1 - Z_2\|_\infty. \tag{1}$$

The entries of the Jacobian $J(Z)$ are $J_{ij} = y(\delta_{ij} - y_j)$, where $y = f(Z)$. The induced L_∞ norm is the maximum absolute row sum. Since $y > 0$ and $\sum_j y_j = 1$, we have:

$$\begin{aligned}\|J(Z)\|_\infty &= \max_i \sum_{j=1}^{d} |J_{ij}| \\ &= \max_i \left(y(1-y) + \sum_{j\neq i} y y_j \right) \\ &= \max_i \left(y(1-y) + y(1-y) \right) \\ &= \max_i 2y(1-y).\end{aligned}$$

The quadratic function $g(x) = 2x(1-x)$ achieves its maximum of $1/2$ at $x = 1/2$. Thus, $\sup_Z \|J(Z)\|_\infty \leq 1/2$. The result follows immediately. □

Lemma 2 (Rademacher Complexity Bound for Vector-Valued Classes [Mohri *et al.*, 2018; Awasthi *et al.*, 2020]). *Let $\{\sigma\}_{i=1}^n$ be a sequence of independent Rademacher random variables, and let $x_1, \ldots, x_n$ be vectors in $\mathbb{R}^d$. Define the matrix $X \in \mathbb{R}^{n\times d}$ with rows x^T. For any $p \in [1, \infty]$ with Hölder conjugate p^* (i.e., $1/p + 1/p^* = 1$), the following bound holds for the expected ℓ_{p^*}-norm of the Rademacher average:*

$$\mathbb{E}_\sigma \left[\left\| \sum_{i=1}^{n} \sigma x \right\|_{p^*} \right] \leq C(p^*) \|X\|_{2,p^*},$$

where the constant $C(p^*)$ and the matrix norm $\|X\|_{2,p^*}$ are given by:

$$C(p^*) = \begin{cases} \sqrt{2\log(2d)}, & \text{if } p = 1 \ (p^* = \infty), \\ \sqrt{2} \left[\frac{\Gamma\left(\frac{1+p^*}{2}\right)}{\sqrt{\pi}} \right]^{1/p^*}, & \text{if } 1 < p \leq 2 \ (2 \leq p^* < \infty), \\ 1, & \text{if } p \geq 2 \ (1 < p^* \leq 2), \end{cases}$$

and $\|X\|_{2,p^*} = \|(\|x_1\|_2, \ldots, \|x_n\|_2)\|_{p^*}$ is the $(2, p^*)$-norm of matrix X.

B Proof of Lemma 4.4

B.1 Proof of lemma 4.1

Let the function class $\mathcal{F} : \mathbb{R}^{\times d} \to \mathbb{R}^K$ be defined in (2). For data i, we define the pairwise class margin between u and v as: $f_{uv}\big(\Theta, \mathbf{X}^i\big) = f_u\big(\Theta, \mathbf{X}^i\big) - f_v\big(\Theta, \mathbf{X}^i\big)$, where $f_u\big(\Theta, \mathbf{X}^i\big)$ represents the predicted score for class u. Our focus is on analyzing how the relative classification error varies between different class pairs. More precisely, we examine the difference in pairwise margins: $f_{uv}(\Theta, \widehat{\mathbf{X}}^i) - f\big(\Theta, \mathbf{X}^i\big)$, which captures the change in model confidence between two class predictions under different input conditions. Then, we have:

$$
\begin{aligned}
& f_{uv}\big(\Theta,\widehat{\mathbf{X}}^i\big)-f\big(\Theta,\mathbf{X}^i\big)\\
\overset{(a)}{=}& \big(\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big)\Big[\mathbf{W}_C^T\sigma\big(\mathbf{W}_V^T\widehat{\mathbf{X}}^{iT}\text{softmax}\big(\widehat{\mathbf{X}}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)\big)\Big]-\big(\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big)\Big[\mathbf{W}_C^T\sigma\big(\mathbf{W}_V^T\mathbf{X}^T\text{softmax}\big(\mathbf{X}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)\big)\Big]\\
\overset{(b)}{\leq}& \big\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big\|_1\big\|\mathbf{W}_C^T\big(\sigma\big(\mathbf{W}_V^T\widehat{\mathbf{X}}^{iT}\text{softmax}\big(\widehat{\mathbf{X}}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)\big)-\sigma\big(\mathbf{W}_V^T\mathbf{X}^T\text{softmax}\big(\mathbf{X}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)\big)\big)\big\|_\infty\\
\overset{(c)}{\leq}& \big\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big\|_1\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\sigma\big(\mathbf{W}_V^T\widehat{\mathbf{X}}^{iT}\text{softmax}\big(\widehat{\mathbf{X}}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)\big)-\sigma\big(\mathbf{W}_V^T\mathbf{X}^T\text{softmax}\big(\mathbf{X}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)\big)\big\|_\infty\\
\overset{(d)}{\leq}& L_\sigma\big\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big\|_1\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\mathbf{W}_V^T\big\|_{1,\infty}\big\|\widehat{\mathbf{X}}^{iT}\text{softmax}\big(\widehat{\mathbf{X}}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)-\mathbf{X}^T\text{softmax}\big(\mathbf{X}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)\big\|_\infty\\
\overset{(e)}{\leq}& L_\sigma\big\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big\|_1\|\mathbf{W}_C^T\|_{1,\infty}\|\mathbf{W}_V^T\|_{1,\infty}\big\|\big(\widehat{\mathbf{X}}^i-\mathbf{X}\big)\text{softmax}\big(\widehat{\mathbf{X}}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)\\
&+\mathbf{X}^i\big[\text{softmax}\big(\widehat{\mathbf{X}}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)-\text{softmax}\big(\mathbf{X}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big)\big]\|_\infty\\
\overset{(f)}{\leq}& L_\sigma\big\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big\|_1\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\mathbf{W}_V^T\big\|_{1,\infty}\big(\epsilon+\frac{\big\|\mathbf{X}\big\|_{1,\infty}}{2}\big\|\big(\widehat{\mathbf{X}}^i-\mathbf{X}\big)\mathbf{W}_{QK}x_{[\mathbf{cls}]}\big\|_\infty\big)\\
\overset{(g)}{=}& L_\sigma\epsilon\big\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big\|_1\|\mathbf{W}_C^T\|_{1,\infty}\|\mathbf{W}_V^T\|_{1,\infty}\big(1+\frac{1}{2}\|\mathbf{W}_{QK}\|_1\|\mathbf{X}\|_{1,\infty}\big\|\mathbf{x}_{[\mathbf{cls}]}\|_\infty\big).
\end{aligned}
$$

Inequality (b) follows from Hölder's inequality: $u^Tv\leq\|u\|_p\|v\|_q$ for $\frac{1}{p}+\frac{1}{q}=1$. Inequality (e) is obtained from the operator norm bound $\|Pv\|_\infty\leq\|P\|_{1,\infty}\|v\|_\infty$. Finally, applying the condition $\|\widehat{\mathbf{X}}^i-\mathbf{X}\|_\infty\leq\epsilon$ together with Lemma 1 yields the subsequent result. In summary we have:

$$
f_{uv}(\Theta,\widehat{\mathbf{X}}^i)-f(\Theta,\mathbf{X}^i)\leq L_\sigma\epsilon\big\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big\|_1\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\mathbf{W}_V^T\big\|_{1,\infty}\big(1+\frac{1}{2}\big\|\mathbf{W}_{QK}\big\|_1\big\|\mathbf{X}\big\|_{1,\infty}\big\|x_{\mathbf{cls}}\big\|_\infty\big) \tag{2}
$$

Then according to the ramp loss and the above inequality, we have:

$$
\begin{aligned}
&\min_{\widetilde{\mathbf{X}}\in\mathcal{B}_{\mathbf{X}}^\infty(\epsilon)}1\big(y^i\neq\arg\max_{y'\in[K]}\big[f\big(\Theta,\widetilde{\mathbf{X}}\big)\big]_{y'}\big)\\
\overset{(a)}{\leq}& l_\gamma\big(\min_{\widetilde{\mathbf{X}}\in\mathcal{B}_{\mathbf{X}}^\infty(\epsilon)}\mathcal{M}\big(f\big(\Theta,\widetilde{\mathbf{X}}\big),y^i\big)\big)\\
\overset{(b)}{\leq}& l_\gamma\big(\min_{y'\neq y^i}\min_{\widetilde{\mathbf{X}}\in\mathcal{B}_{\mathbf{X}}^\infty(\epsilon)}\big[f\big(\Theta,\widetilde{\mathbf{X}}\big)\big]_{y^i}-\big[f\big(\Theta,\widetilde{\mathbf{X}}\big)\big]_{y'}\big)\\
\overset{(c)}{\leq}& l_\gamma\big(\min_{y'\neq y^i}\big[f\big(\Theta,\mathbf{X}\big)\big]_{y^i}-\big[f\big(\Theta,\mathbf{X}\big)\big]_{y'}\\
&-\max_{y'\neq y^i}L_\sigma\epsilon\big(\frac{\big\|\mathbf{X}\big\|_{1,\infty}\big\|x_{\mathbf{cls}}\big\|_\infty\big\|\mathbf{W}_{QK}\big\|_1}{2}+1\big)\big\|w_{*u}^T-w_{*v}^T\big\|_1\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\mathbf{W}_V^T\big\|_{1,\infty}\big)\\
\overset{(d)}{\leq}& l_\gamma\big(M\big(f\big(\Theta,\mathbf{X}\big),y^i\big)-2\max_{k\in[K]}L_\sigma\epsilon\big(\frac{\|\mathbf{X}\|_{1,\infty}\big\|x_{\mathbf{cls}}\big\|_\infty\|\mathbf{W}_{QK}\|_1}{2}+1\big)\|w_{k*}^T\|_1\|\mathbf{W}_C^T\|_{1,\infty}\|\mathbf{W}_V^T\|_{1,\infty}\big)\\
\overset{(e)}{\leq}& 1\big(M\big(f\big(\Theta,\mathbf{X}\big),y^i\big)-2\max_{k\in[K]}L_\sigma\epsilon\big(\frac{\|\mathbf{X}\|_{1,\infty}\big\|x_{\mathbf{cls}}\big\|_\infty\|\mathbf{W}_{QK}\|_1}{2}+1\big)\|w_{k*}^T\|_1\|\mathbf{W}_C^T\|_{1,\infty}\|\mathbf{W}_V^T\|_{1,\infty}\leq\gamma\big).
\end{aligned}
$$

Inequality (a) follows from the property of the ramp loss, inequality (b) comes from the definition of the margin operator, inequality (c) is derived from inequality (2), inequality (d) uses the triangle inequality property, and inequality (e) directly follows from the property of the ramp loss. Theorem 4.1 establishes the foundation for subsequent generalization bound analysis (Theorem 4.4) by constructing a robust surrogate loss function that quantifies inter-class margin variations in ViTs under adversarial perturbations.

B.2 Proof of lemma 4.4

By the Ledoux-Talagrand contraction inequality, we know that

$$\mathfrak{R}_n\big(\widetilde{\ell}\circ\mathcal{F}\big) \leq \mathfrak{R}_n\big(\widehat{\ell}\circ\mathcal{F}\big) \leq \frac{1}{\gamma}\big(\mathfrak{R}_n\big(M\circ\mathcal{F}\big) + \mathfrak{R}_n\big(\Lambda\circ\mathcal{F}\big)\big).$$

For the right side of the above inequality, $\mathfrak{R}_n\big(\Lambda\circ\mathcal{F}\big)$ can be bound by,

$$\begin{aligned}
\mathfrak{R}_n\big(\Lambda\circ\mathcal{F}\big) &\leq 2\max_{k\in[K]} L_\sigma\epsilon\big\|w_{k*}^T\big\|_1\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\mathbf{W}_V^T\big\|_{1,\infty}\mathbb{E}_\sigma\left|\sup\frac{1}{n}\sum_{i=1}^n\sigma\left(\frac{\|\mathbf{X}^i\|_{1,\infty}\big\|x_{\mathbf{cls}}^i\big\|_\infty\|\mathbf{W}_{QK}\|_1}{2}+1\right)\right| \\
&\leq 2KL_\sigma\epsilon\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\mathbf{W}_V^T\big\|_{1,\infty}\left[\mathbb{E}_\sigma\left|\sup\frac{1}{n}\sum_{i=1}^n\sigma\left(\frac{\|\mathbf{X}^i\|_{1,\infty}\big\|x_{\mathbf{cls}}^i\big\|_\infty\big\|\mathbf{W}_{QK}\big\|_1}{2}\right)\right|+\mathbb{E}_\sigma\left|\sup\frac{1}{n}\sum_{i=1}^n\sigma_i\right|\right] \\
&\leq Kn^{-1/2}L_\sigma\epsilon\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\mathbf{W}_V^T\big\|_{1,\infty}\big(\|\mathbf{X}\|_{1,\infty}\big\|\mathbf{W}_{QK}\big\|_1\mathbb{E}_\sigma\left|\sup\sum_{i=1}^n\big\|\sigma_i\mathbf{x}_{cls}^i\big\|_\infty\right|+2\big) \\
&\leq Kn^{-1/2}L_\sigma\epsilon\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\mathbf{W}_V^T\big\|_{1,\infty}\big(\sqrt{2\log(2d)}\|\mathbf{X}\|_{1,\infty}\big\|\mathbf{W}_{QK}\big\|_1\big\|\mathbf{X}\big\|_{2,\infty}+2\big).
\end{aligned}$$

For the left side of the inequality, $\mathfrak{R}_n\big(M\circ\mathcal{F}\big)$ can be bound by ,

$$M\big(f(\Theta,\mathbf{X}),y\big) = \mathbf{w}_{y^i}^T Y_{cls} - \max_{j\neq y^i}\mathbf{w}_j^T Y_{cls}.$$

In the computation of Rademacher complexity, the maximization operation in the $\mathcal{M}$ operator can be relaxed to a linear combination across all classes. This relaxation is achievable through norm control of the classifier parameters, which provides an implicit bound on the margin $\mathcal{M}$.

$$\begin{aligned}
&\mathfrak{R}_S\big(M\circ\mathcal{F}\big) = \mathbb{E}_\sigma\big[\sup\frac{1}{n}\sum_{i=1}^n\sigma_i M\big(f(\Theta,\mathbf{X}^i),y^i\big)\big] \\
=&\mathbb{E}_\sigma\big[\sup\frac{1}{n}\sum_{i=1}^n\langle\sigma_i,\mathbf{w}_{*y^i}^T\mathbf{W}_C^T\sigma\big(\mathbf{W}_V^T\mathbf{X}^T\mathrm{softmax}\big(\mathbf{X}\mathbf{W}_{QK}x_{[\mathbf{cls}]}^i\big)\big)\rangle\big] \\
\leq& n^{-1/2}L_\sigma\big\|\mathbf{w}^T\big\|_2\big\|\mathbf{W}_C^T\big\|_2\big\|\mathbf{W}_V^T\big\|_2\mathbb{E}_\sigma\left[\sup\sum_{i=1}^n\left\|\sigma_i\mathbf{X}^{iT}\mathrm{softmax}\big(\mathbf{X}^i\mathbf{W}_{QK}x_{[\mathbf{cls}]}^i\big)\right\|_2\right] \\
\leq& n^{-1/2}L_\sigma\big\|\mathbf{w}^T\big\|_2\big\|\mathbf{W}_C^T\big\|_2\big\|\mathbf{W}_V^T\big\|_2\big\|\mathbf{X}\big\|_2^2\big\|\mathbf{W}_{QK}\big\|_1\mathbb{E}_\sigma\big[\sup\sum_{i=1}^n\big\|\sigma x_{cls}^i\big\|_2\big] \\
\leq& Cn^{-1/2}L_\sigma\big\|\mathbf{w}^T\big\|_2\big\|\mathbf{W}_C^T\big\|_2\big\|\mathbf{W}_V^T\big\|_2\big\|\mathbf{X}\big\|_2^2\big\|\mathbf{W}_{QK}\big\|_1\|\mathbf{X}\|_{2,2},
\end{aligned}$$

where $C=\sqrt{2}[\frac{\Gamma(3/2)}{\sqrt{\pi}}]^{1/2}$ is a constant. Adding the $\mathfrak{R}_n\big(M\circ\mathcal{F}\big)$ and $\mathfrak{R}_n\big(\Lambda\circ\mathcal{F}\big)$ we final get the bound of $\mathfrak{R}_n\big(\widetilde{\ell}\circ\mathcal{F}\big)$,

$$\begin{aligned}
&\mathfrak{R}_n\big(\widetilde{\ell}\circ\mathcal{F}\big) \\
\leq& Kn^{-1/2}\gamma^{-1}L_\sigma\epsilon\big\|\mathbf{W}_C^T\big\|_{1,\infty}\big\|\mathbf{W}_V^T\big\|_{1,\infty}\big(\sqrt{2\log(2d)}\|\mathbf{X}\|_{1,\infty}\big\|\mathbf{W}_{QK}\big\|_1\big\|\mathbf{X}\big\|_{2,\infty}+2\big)+C\gamma^{-1}n^{-1/2}L_\sigma\big\|\mathbf{w}^T\big\|_2\big\|\mathbf{W}_C^T\big\|_2\big\|\mathbf{W}_V^T\big\|_2 \\
\leq& \gamma^{-1}n^{-1/2}L_\sigma\mathbf{B}^2\big\|\mathbf{W}_{QK}\big\|_1\big(K\epsilon\sqrt{2\log(2d)}\|\mathbf{X}\|_{1,\infty}\big\|\mathbf{X}\big\|_{2,\infty}+C\big\|\mathbf{w}^T\big\|_2\|\mathbf{X}\|_2^2\big\|\mathbf{X}\big\|_{2,2}+2\big).
\end{aligned}$$

Knowing that $\|X\|_{2,\infty} = \max_i \|x\|_2$, $\|X\|_{1,\infty} = \max_i \|x\|_1$ and $\|X\|_{2,2} \leq \|X\|_F \leq \sqrt{M}\max_i(\|x\|_2)$. Therefore

$$\begin{aligned}\mathfrak{R}_n\big(\widetilde{\ell}\circ\mathcal{F}\big) \leq & \gamma^{-1}n^{-1/2}L_\sigma\mathbf{B}^2\big\|\mathbf{W}_{QK}\big\|_1\big(K\epsilon\sqrt{2\log(2d)}(\max\big\|\mathbf{X}^i\big\|_2^2) + CM^{3/2}\big\|\mathbf{w}^T(\max\big\|\mathbf{x}\big\|_2^{3/2}) + 2\big)\\ \leq & \gamma^{-1}n^{-1/2}L_\sigma\mathbf{B}^2\big\|\mathbf{W}_{QK}\big\|_1\big(K\epsilon\sqrt{2\log(2d)}\mathbf{B}_X^2 + CM^{3/2}\big\|\mathbf{w}^T\big\|\mathbf{B}_X^3 + 2\big).\end{aligned}$$

with $C = \sqrt{2}[\frac{\Gamma(3/2)}{\sqrt{\pi}}]^{1/2}$ and $\mathbf{B}_X = \max_i \|\mathbf{x}_i\|_2$.

C Proof of Lemma 4.6

In this section we will systematically analyze the behavior of deep ViTs under adversarial perturbations by constructing feature propagation upper bounds $g_{block}^{(l+1)}\big(\mathbf{W}^{1:L+1};\mathbf{X}\big)$ and $g_{block}^{(l+1)}\big(\mathbf{W}^{1:L+1};\widetilde{\mathbf{X}}\big)$.

$$\begin{aligned}&\big\|g_{block}^{(l+1)}\big(\mathbf{W}^{1:L+1};\mathbf{X}\big)\big\|_2\\ =&\big\|\mathbf{W}_C^{(l)}\sigma\big(\mathbf{W}_V^{(l)}g_{block}^{(l)}\text{softmax}\big(g_{block}^{(l)}\mathbf{W}_{QK}^{(l)T}g_{block}^{(l)}\big)\big)\big\|_2\\ \leq& L_\sigma\big\|\mathbf{W}_C^{(l)}\big\|_2\big\|\mathbf{W}_V^{(l)}\big\|_2\big\|g_{block}^{(l)}\big(\mathbf{W}^{1:L};\mathbf{X}\big)\big\|_2\\ \leq& L_\sigma^{(l)}\big\|\mathbf{X}\big\|_2\prod_{l=1}^{L}\big(\big\|\mathbf{W}_C^{(l)}\big\|_2\big\|\mathbf{W}_V^{(l)}\big\|_2\big).\end{aligned}$$

Then we denote $\Psi_{L+1} = L_\sigma^{(L)}\big\|\mathbf{X}\big\|_2\prod_{l=1}^{L}\big(\big\|\mathbf{W}_C^{(l)}\big\|_2\big\|\mathbf{W}_V^{(l)}\big\|_2\big)$.

$$\begin{aligned}&\big\|g_{block}^{(l+1)}\big(\mathbf{W}^{1:L+1};\widetilde{\mathbf{X}}\big)\big\|_2\\ =&\ \big\|\mathbf{W}_C^{(l)}\sigma\big(\mathbf{W}_V^{(l)}g_{block}^{(l)}\text{softmax}\big(g_{block}^{(l)}\mathbf{W}_{QK}^{(l)T}g_{block}^{(l)}\big)\big)\big\|_2\\ \leq&\ L_\sigma\big\|\mathbf{W}_C^{(l)}\big\|_2\big\|\mathbf{W}_V^{(l)}\big\|_2\big\|g_{block}^{(l)}\big(\mathbf{W}^{1:L};\widetilde{\mathbf{X}}\big)\big\|_2\\ \leq&\ L_\sigma^{(l)}\big\|\widetilde{\mathbf{X}}\big\|_2\prod_{l=1}^{L}\big(\big\|\mathbf{W}_C^{(l)}\big\|_2\big\|\mathbf{W}_V^{(l)}\big\|_2\big).\end{aligned}$$

Similarly we denote $\Psi'_{L+1} = L_\sigma^{(L)}\big(\big\|\mathbf{X}\big\|_2+\epsilon\big)\prod_{l=1}^{L}\big(\big\|\mathbf{W}_C^{(l)}\big\|_2\big\|\mathbf{W}_V^{(l)}\big\|_2\big)$.

These two critical quantities explicitly capture the multiplicative effects of inter-layer weight matrices $\prod_{l=1}^{L}\big(\big\|\mathbf{W}_C^{(l)}\big\|_2\big\|\mathbf{W}_V^{(l)}\big\|_2\big)$ and the cumulative scaling of Lipschitz activation functions L_σ^L. Here, Ψ_{L+1} rigorously constrains the propagation upper bound of the original input $\mathbf{X}$'s L_2-norm across the Transformer's hierarchical structure, while Ψ'_{L+1}extends the norm-based analysis framework to adversarial samples X by incorporating the perturbation intensity ϵ. This construction provides a solid foundation for our subsequent proof of Theorem 4.9, significantly simplifying the theoretical derivation.

Let the function class $\mathcal{G}_{scalar}^{(l)}: \mathbb{R}^{m\times d}\to\mathbb{R}^K$ be defined in (3). Similarly, let the pairwise class margin of image i be defined as:

$$\mathcal{G}_{uv}^{(l)}\big(\mathbf{X}^i;\mathbf{W}^{1:L+1},w\big) = \big[\mathcal{G}^{(l)}\big(\mathbf{X}^i;\mathbf{W}^{1:L+1},w\big)\big]_u - \big[\mathcal{G}^{(l)}\big(\mathbf{X}^i;\mathbf{W}^{1:L+1},w\big)\big]_v,$$

where$\big[\mathcal{G}^{(l)}\big(\mathbf{X}^i;\mathbf{W}^{1:L+1},w\big)\big]_u$ denotes the prediction score for class u at image i.

We would like to observe the relative change in error between any two classes. Specifically, we consider the difference between the set of pairwise margins:

$$\mathcal{G}_{uv}^{(l)}\big(\widetilde{\mathbf{X}}^i;\mathbf{W}^{1:L+1},w\big) - \mathcal{G}_{uv}^{(l)}\big(\mathbf{X}^i;\mathbf{W}^{1:L+1},w\big).$$

Define $g_{block}^{(l)}(\widetilde{\mathbf{X}};\mathbf{W}^{1:L})$ and $g_{block}^{(l)}(\mathbf{X};\mathbf{W}^{1:L})$ as the feature representation of $\widetilde{\mathbf{X}}$ and $\mathbf{X}$ at the l-th layer, respectively.

$$
\begin{aligned}
&\mathcal{G}_{uv}^{(l)}(\widetilde{\mathbf{X}};\mathbf{W}^{1:L+1},w)-\mathcal{G}_{uv}^{(l)}(\mathbf{X}^i;\mathbf{W}^{1:L+1},w)\\
&=(\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T)g_{block}^{(l+1)}(\widetilde{\mathbf{X}}^i;\mathbf{W}^{1:L+1},w)-(\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T)g_{block}^{(l+1)}(\mathbf{X}^i;\mathbf{W}^{1:L+1},w)\\
&=(\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T)\big[\prod\nolimits_{norm}\big(\sigma\big(\prod\nolimits_{norm}\big(f\big(g_{block}^{(l)}(\widetilde{\mathbf{X}}^i;\mathbf{W}^{1:L}),\mathbf{W}^{(l)}\big)\big)\big)\mathbf{W}_C^{(l)}\big)\big]\\
&-(\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T)\big[\prod\nolimits_{norm}\big(\sigma\big(\prod\nolimits_{norm}\big(f\big(g_{block}^{(l)}(\mathbf{X}^i;\mathbf{W}^{1:L}),\mathbf{W}^{(l)}\big)\big)\big)\mathbf{W}_C^{(l)}\big)\big]\\
&\leq\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\|_1\|\sigma\big(\prod\nolimits_{norm}\big(f\big(g_{block}^{(l)}(\widetilde{\mathbf{X}}^i;\mathbf{W}^{1:L}),\mathbf{W}^{(l)}\big)\big)\big)\mathbf{W}_C^{(l)}-\sigma\big(\prod\nolimits_{norm}\big(f\big(g_{block}^{(l)}(\mathbf{X}^i;\mathbf{W}^{1:L}),\mathbf{W}^{(l)}\big)\big)\big)\mathbf{W}_C^{(l)}\|_2\\
&\leq L_\sigma\|\mathbf{W}_C^{(l)}\|_2\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\|_1\|f\big(g_{block}^{(l)}(\widetilde{\mathbf{X}}^i;\mathbf{W}^{1:L}),\mathbf{W}^{(l)}\big)-f\big(g_{block}^{(l)}(\mathbf{X}^i;\mathbf{W}^{1:L}),\mathbf{W}^{(l)}\big)\|_2.
\end{aligned}
$$

To facilitate analysis,We denote $g_{block}^{(l)}(\widetilde{\mathbf{X}}^i;\mathbf{W}^{1:L}),g_{block}^{(l)}(\mathbf{X}^i;\mathbf{W}^{1:L})$as $\widetilde{g}_{block}^{(l)}$ and $g_{block}^{(l)}$, respectively.

$$
\begin{aligned}
&\|f(\widetilde{g}_{block}^{(l)},\mathbf{W}^{(l)})-f(g_{block}^{(l)},\mathbf{W}^{(l)})\|_2\\
=&\|\mathbf{W}_V^{(l)T}\widetilde{g}_{block}^{(l)}\text{softmax}(\widetilde{g}_{block}^{(l)}\mathbf{W}_{QK}^{(l)}\widetilde{g}_{block}^{(l)})-\mathbf{W}_V^{(l)T}g_{block}^{(l)}\text{softmax}(g_{block}^{(l)}\mathbf{W}_{QK}^{(l)}g_{block}^{(l)})\|_2\\
\leq&\|\mathbf{W}_V^{(l)T}[\widetilde{g}_{block}^{(l)}-g_{block}^{(l)}]\text{softmax}(\widetilde{g}_{block}^{(l)}\mathbf{W}_{QK}^{(l)}\widetilde{g}_{block}^{(l)})\|_2\\
+&\|\mathbf{W}_V^{(l)T}g_{block}^{(l)}[\text{softmax}(\widetilde{g}_{block}^{(l)}\mathbf{W}_{QK}^{(l)}\widetilde{g}_{block}^{(l)})-\text{softmax}(g_{block}^{(l)}\mathbf{W}_{QK}^{(l)}g_{block}^{(l)})]\|_2\\
\leq&\|\mathbf{W}_V^{(l)T}\|_2\|\widetilde{g}_{block}^{(l)}-g_{block}^{(l)}\|_2+L_{\text{softmax}}\|\mathbf{W}_V^{(l)T}\|_2\|\mathbf{W}_{QK}^{(l)T}\|_2\|g_{block}^{(l)}\|_2\|\widetilde{g}_{block}^{(l)}-g_{block}^{(l)}\|_2(\|\widetilde{g}_{block}^{(l)}\|_2+\|g_{block}^{(l)}\|_2)\\
=&\|\mathbf{W}_V^{(l)T}\|_2\|\widetilde{g}_{block}^{(l)}-g_{block}^{(l)}\|_2(1+L_{\text{softmax}}\|\mathbf{W}_{QK}^{(l)T}\|_2\|g_{block}^{(l)}\|_2(\|\widetilde{g}_{block}^{(l)}\|_2+\|g_{block}^{(l)}\|_2))\\
\leq&\|\mathbf{W}_V^{(l)T}\|_2\|\widetilde{g}_{block}^{(l)}-g_{block}^{(l)}\|_2[1+\|\mathbf{W}_{QK}^{(l)}\|_2\Psi_{L+1}(\Psi_{L+1}+\Psi'_{L+1})].
\end{aligned}
$$

To generalize across layers, we recursively apply the bound to earlier layers:

$$
\begin{aligned}
&\|\widetilde{g}_{block}^{(l)}-g_{block}^{(l)}\|_2\\
=&\Big\|\prod_{norm}\big(\sigma\big(\prod_{norm}\big(f\big(g_{block}^{(L-1)}(\mathbf{W}^{1:L-1};\widetilde{\mathbf{X}}^i),\mathbf{W}^{(L-1)}\big)\big)\big)\mathbf{W}_C^{(L-1)}\big)\\
-&\prod_{norm}\big(\sigma\big(\prod_{norm}\big(f\big(g_{block}^{(L-1)}(\mathbf{W}^{1:L-1};\mathbf{X}^i),\mathbf{W}^{(L-1)}\big)\big)\big)\mathbf{W}_C^{(L-1)}\big)\Big\|_2\\
\leq&L_\sigma\|\mathbf{W}_C^{(L-1)}\|_2\|f\big(g_{block}^{(L-1)}(\mathbf{W}^{1:L-1};\widetilde{\mathbf{X}}^i),\mathbf{W}^{(L-1)}\big)-f\big(g_{block}^{(L-1)}(\mathbf{W}^{1:L-1};\mathbf{X}^i),\mathbf{W}^{(L-1)}\big)\|_2.
\end{aligned}
$$

By unrolling the recursion, we obtain a product of layer-wise terms:

$$
\begin{aligned}
&\|f(\widetilde{g}_{block}^{(l)},\mathbf{W}^{(l)})-f(g_{block}^{(l)},\mathbf{W}^{(l)})\|_2\\
\leq&L_\sigma\|\mathbf{W}_C^{(L-1)}\|_2\|\mathbf{W}_V^{(l)}\|_2[1+\|\mathbf{W}_{QK}^{(l)}\|_2\Psi_{l+1}(\Psi_{l+1}+\Psi'_{l+1})]\\
\times&\|f\big(g_{block}^{(L-1)}(\mathbf{W}^{1:L-1};\widetilde{\mathbf{X}}^i),\mathbf{W}^{(L-1)}\big)-f\big(g_{block}^{(L-1)}(\mathbf{W}^{1:L-1};\mathbf{X}^i),\mathbf{W}^{(L-1)}\big)\|_2\\
\leq&L_\sigma^{L-1}\prod\nolimits_{l=1}^{L}[1+\|\mathbf{W}_{QK}^{(l)}\|_2\Psi_{l+1}(\Psi_{l+1}+\Psi'_{l+1})]\|\mathbf{W}_V^{(l)}\|_2\prod\nolimits_{j=1}^{(L-1)}\|\mathbf{W}_C^{(j)}\|_2.
\end{aligned}
$$

Combining all steps, the change in pairwise margin is bounded by:

$$
\begin{aligned}
&\mathcal{G}_{uv}^{(L)}\big(\widetilde{\mathbf{X}}^i;\mathbf{W}^{1:L+1},w\big)-\mathcal{G}_{uv}^{(L)}\big(\mathbf{X}^i;\mathbf{W}^{1:L+1},w\big)\\
\leq& L_\sigma^L\big\|\mathbf{w}_{*u}^T-\mathbf{w}_{*v}^T\big\|_1\prod\nolimits_{l=1}^{L}\big[\big\|\mathbf{W}_V^{(l)}\big\|_2\big\|\mathbf{W}_C^{(l)}\big\|_2\big(1+\big\|\mathbf{W}_{QK}^{(l)}\big\|_2\Psi_{L+1}\big(\Psi_{L+1}+\Psi'_{L+1}\big)\big)\big]\\
\leq& L_\sigma^L\max_{k\in[K]}\big\|\mathbf{w}_{*k}^T\big\|_1\prod\nolimits_{l=1}^{L}\big[\big\|\mathbf{W}_V^{(l)}\big\|_2\big\|\mathbf{W}_C^{(l)}\big\|_2\big(1+\big\|\mathbf{W}_{QK}^{(l)}\big\|_2\Psi_{L+1}\big(\Psi_{L+1}+\Psi'_{L+1}\big)\big)\big].
\end{aligned}
\tag{3}
$$

Finally we denote (3) as $\Lambda\big(\mathcal{G}\big(\mathbf{X}^i;\mathbf{W}^{1:L},w\big)\big)$.

C.1 Proof of Lemma 4.6

Since $\mathfrak{R}_n\big(\widetilde{\ell}\circ\mathcal{G}_{scalar}^{(l)}\big)\leq\mathfrak{R}_n\big(\widehat{\ell}\circ\mathcal{G}_{scalar}^{(l)}\big)\leq\frac{1}{\gamma}\big(\mathfrak{R}_n\big(M\circ\mathcal{G}_{scalar}^{(l)}\big)+\mathfrak{R}_n\big(\Lambda\circ\mathcal{G}_{scalar}^{(l)}\big)\big)$.
The right side of the above equation can be bound by

$$
\begin{aligned}
&\mathfrak{R}_n\big(\Lambda\circ\mathcal{G}_{scalar}^{(l)}\big)\\
\leq& 2L_\sigma^L\max_{k\in[K]}\big\|\mathbf{w}_{*k}^T\big\|_1\prod\nolimits_{l=2}^{L}\big[\big\|\mathbf{W}_V^{(l)}\big\|_2\big\|\mathbf{W}_C^{(l)}\big\|_2\big(1+\big\|\mathbf{W}_{QK}^{(l)}\big\|_2\Psi_{l+1}\big(\Psi_{l+1}+\Psi'_{l+1}\big)\big)\big]\\
&\times\mathbb{E}_\sigma\left|\sup\frac{1}{n}\sum_{i=1}^{n}\sigma_i\mathbf{W}_C\big(\mathbf{W}_V\mathbf{X}^i\mathrm{softmax}(\mathbf{X}^i\mathbf{W}_{QK}\mathbf{x}_{[\mathbf{cls}]})\big)\right|\\
\leq& 2Kn^{-1/2}L_\sigma^L\prod\nolimits_{l=2}^{L}\big[\big\|\mathbf{W}_V^{(l)}\big\|_2\big\|\mathbf{W}_C^{(l)}\big\|_2\big(1+\big\|\mathbf{W}_{QK}^{(l)}\big\|_2\Psi_{l+1}\big(\Psi_{l+1}+\Psi'_{l+1}\big)\big)\big]\\
&\times\mathbb{E}_\sigma\left[\sup\sum_{i=1}^{n}\left\|\sigma_i\mathbf{W}_C\big(\mathbf{W}_V\mathbf{X}^i\mathrm{softmax}(\mathbf{X}^i\mathbf{W}_{QK}\mathbf{x}_{[\mathbf{cls}]})\right\|\big)\right]\\
\leq& 2Kn^{-1/2}L_\sigma^L\big\|\mathbf{W}_C\big\|_2\prod\nolimits_{l=2}^{L}\big[\big\|\mathbf{W}_V^{(l)}\big\|_2\big\|\mathbf{W}_C^{(l)}\big\|_2\big(1+\big\|\mathbf{W}_{QK}^{(l)}\big\|_2\Psi_{l+1}\big(\Psi_{l+1}+\Psi'_{l+1}\big)\big)\big]\mathbb{E}_\sigma\left[\sup\sum_{i=1}^{n}\left\|\sigma_i\mathbf{W}_V\mathbf{x}_{cls}^i\right\|\right]\\
\leq& 2Kn^{-1/2}L_\sigma^L\big\|\mathbf{W}_C\big\|_2\big\|\mathbf{W}_V\big\|_1\prod\nolimits_{l=2}^{L}\big[\big\|\mathbf{W}_V^{(l)}\big\|_2\big\|\mathbf{W}_C^{(l)}\big\|_2\big(1+\big\|\mathbf{W}_{QK}^{(l)}\big\|_2\Psi_{l+1}\big(\Psi_{l+1}+\Psi'_{l+1}\big)\big)\big]\mathbb{E}_\sigma\left[\sup\sum_{i=1}^{n}\left\|\sigma_i\mathbf{x}_{cls}^i\right\|_\infty\right]\\
\leq& 2K\sqrt{2\log(2d)}n^{-1/2}L_\sigma^L\big\|\mathbf{W}_C\big\|_2\big\|\mathbf{W}_V\big\|_1\prod\nolimits_{l=2}^{L}\big[\big\|\mathbf{W}_V^{(l)}\big\|_2\big\|\mathbf{W}_C^{(l)}\big\|_2\big(1+\big\|\mathbf{W}_{QK}^{(l)}\big\|_2\Psi_{l+1}\big(\Psi_{l+1}+\Psi'_{l+1}\big)\big)\big]\big\|\mathbf{X}\big\|_{2,\infty}
\end{aligned}
\tag{4}
$$

Meanwhile, the $\mathfrak{R}_n\big(M\circ\mathcal{G}_{scalar}^{(l)}\big)$ can be bound by

$$
\begin{aligned}
&\mathfrak{R}_n\big(M\circ\mathcal{G}_{scalar}^{(l)}\big)\\
=&\mathbb{E}_\sigma\left[\sup_\sigma\frac{1}{n}\sum_{i=1}^{n}\langle\sigma_i,\big(\mathbf{w}^T\big[g\big(\mathbf{X}^i,\mathbf{W}^{1:L+1}\big)\big]\big)\rangle\right]\\
\leq& n^{-1/2}L_\sigma\big\|\mathbf{w}^T\big\|_2\big\|\mathbf{W}_C^{(l)}\big\|_2\big\|\mathbf{W}_V^{(l)}\big\|_2\mathbb{E}_\sigma\left[\sup_\sigma\frac{1}{n}\sum_{i=1}^{n}\left\|\sigma_i g\big(\mathbf{X}^i,\mathbf{W}^{1:L}\big)\right\|\right]\\
\leq& n^{-1/2}L_\sigma^L\big\|\mathbf{w}^T\big\|_2\prod_{l=1}^{L}\big\|\mathbf{W}_C^{(l)}\big\|_2\prod_{l=2}^{L}\big\|\mathbf{W}_V^{(l)}\big\|_2\,\mathbb{E}_\sigma\left[\sup_\sigma\frac{1}{n}\sum_{i=1}^{n}\left\|\sigma_i\mathbf{W}_V\mathbf{X}^i\right\|\right]\\
\leq& n^{-1/2}L_\sigma^L\big\|\mathbf{w}^T\big\|_2\prod_{l=1}^{L}\big(\big\|\mathbf{W}_C^{(l)}\big\|_2\big\|\mathbf{W}_V^{(l)}\big\|_2\big)\big\|\mathbf{X}^i\big\|.
\end{aligned}
\tag{5}
$$

We add (4) and (5) can get the $\mathfrak{R}_n\big(\widetilde{\ell}\circ\mathcal{G}_{scalar}^{(l)}\big)$.

D Future Work

While this study establishes a theoretical framework for the adversarial generalization of ViTs and experimentally validates its core findings, several avenues warrant further exploration. Future research could leverage alternative theoretical lenses, such as algorithmic stability [1, 2] or information-theoretic frameworks [3, 4], to derive tighter generalization bounds or offer complementary perspectives. Moreover, as the current analysis is confined to supervised learning, extending this inquiry to large-scale pre-trained ViT models represents a vital direction. Given the dominance of pre-trained models in transfer learning, analyzing their robust generalization in downstream tasks carries significant theoretical and practical weight.

E Additional Experiment Result

Supplementary to the experimental analysis in the main text, Tables 2 through 5 present detailed numerical results quantifying the impact of input image resolution and adversarial perturbation strength on the generalization performance of ViTs. Consistent with the experimental setup detailed in Section 5, we employed the ViT-Base architecture pre-trained on ImageNet-21k and fine-tuned it using PGD adversarial training across four benchmark datasets: Flower-102, Oxford-IIIT Pets, CIFAR-10, and CIFAR-100.

To empirically validate our theoretical insights regarding the relationship between sequence length (M) and adversarial robustness, we evaluated model performance across varying spatial resolutions. Given that ViTs process images by flattening fixed-size patches (set to 16×16), varying the input resolution results in input sequences with a fixed token dimension ($d = 768$) but differing sequence lengths (M). Specifically, for CIFAR-10/100 (originally 32×32), images were upsampled to resolutions of 224×224, 240×240, and 256×256, corresponding to sequence lengths of $M = 196$, 225, and 256, respectively. Conversely, for the fine-grained Flower-102 and Pets-37 datasets, images were resized to 384×384, 400×400, and 416×416, resulting in sequence lengths of $M = 576$, 625, and 676. Tables 2 and 3 report the accuracies for Flower-102 and Pets-37, while Tables 4 and 5 detail the results for CIFAR-10 and CIFAR-100. For all experiments, we assessed the models against a range of adversarial perturbation budgets $\epsilon \in \{0, 2/255, 4/255, 6/255, 8/255, 10/255\}$, demonstrating the interplay between suquence length and adversarial generalization error.

To provide further empirical validation for the regularization effect of attention sparsity, we present detailed results from ablation studies in Tables 6 through 9 for the Flower-102, Pets-37, CIFAR-10, and CIFAR-100 datasets. These tables systematically report the training accuracy and testing accuracy of the ViT model when subjected to the Drop-Key mechanism with varying ratios ρ and under different adversarial attack strengths ϵ.

Flower-102	Image Size	ϵ					
		0	2/255	4/255	6/255	8/255	10/255
Training	576	100.00±0.00	99.90±0.00	100.00±0.00	99.90±0.00	99.90±0.00	99.90±0.00
	400	99.22±0.00	99.41±0.00	99.41±0.00	99.31±0.00	99.22±0.00	99.12±0.00
	676	99.22±0.00	99.51±0.00	99.51±0.00	99.51±0.00	99.38±0.05	99.22±0.00
Testing	576	99.22±0.00	98.20±0.05	97.84±0.00	97.52±0.05	97.29±0.12	96.96±0.00
	400	97.65±0.00	97.22±0.09	96.73±0.05	96.18±0.00	95.78±0.00	95.78±0.00
	676	97.65±0.00	96.99±0.05	96.67±0.00	96.37±0.00	96.18±0.08	95.88±0.00

Table 2: Adversarial training and testing accuracy on **Flower-102** dataset (mean ± standard deviation, %) under different image sizes and adversarial attack strengths ϵ.

Pets-37	Image Size	ϵ					
		0	2/255	4/255	6/255	8/255	10/255
Train	576	100.00±0.00	99.97±0.00	99.84±0.00	99.70±0.01	99.65±0.01	99.60±0.01
	400	100.00±0.00	99.97±0.00	99.95±0.00	99.92±0.00	99.77±0.01	99.49±0.01
	676	100.00±0.00	99.95±0.01	99.94±0.01	99.82±0.01	99.78±0.00	99.48±0.02
Test	576	96.09±0.00	87.62±0.03	85.39±0.02	84.83±0.05	84.17±0.06	82.18±0.09
	400	94.22±0.00	87.29±0.03	86.40±0.06	83.90±0.06	82.46±0.05	81.34±0.05
	676	94.22±0.00	87.16±0.08	85.25±0.12	83.35±0.04	82.12±0.02	81.01±0.01

Table 3: Adversarial training and testing accuracy on **Pets-37** dataset (mean ± standard deviation, %) under different image sizes and adversarial attack strengths ϵ.

Cifar-10	Image Size	ϵ					
		0	2/255	4/255	6/255	8/255	10/255
Train	196	100.00±0.06	99.98±0.00	99.96±0.00	99.98±0.05	99.95±0.01	99.96±0.00
	225	100.00±0.00	100.00±0.00	100.00±0.00	99.95±0.02	99.96±0.00	99.84±0.00
	256	100.00±0.00	100.00±0.00	99.94±0.00	99.90±0.00	99.90±0.00	99.78±0.00
Test	196	99.00±0.00	94.07±0.12	93.03±0.05	92.50±0.00	91.70±0.08	91.57±0.05
	225	99.30±0.00	93.67±0.05	92.63±0.09	91.57±0.05	91.03±0.05	90.47±0.05
	256	99.00±0.00	93.00±0.14	92.07±0.09	91.27±0.05	90.30±0.08	90.03±0.05

Table 4: Adversarial training and testing accuracy on **CIFAR-10** dataset (mean ± standard deviation, %) under different image sizes and adversarial attack strengths ϵ.

Cifar-100	Image Size	ϵ					
		0	2/255	4/255	6/255	8/255	10/255
Train	196	100.00±0.00	99.66±0.02	99.52±0.03	99.33±0.01	99.17±0.01	99.04±0.02
	225	100.00±0.00	99.56±0.03	99.47±0.01	99.31±0.01	99.04±0.13	98.95±0.01
	256	99.98±0.00	99.52±0.02	99.43±0.04	99.31±0.02	99.10±0.03	98.73±0.17
Test	196	93.80±0.00	79.53±0.17	77.80±0.16	76.77±0.09	74.70±0.08	74.13±0.17
	225	94.00±0.00	78.23±0.05	76.93±0.05	74.83±0.21	73.87±0.26	73.63±0.12
	256	93.60±0.00	77.30±0.14	74.63±0.05	72.67±0.19	72.10±0.08	71.07±0.21

Table 5: Adversarial training and testing accuracy on **CIFAR-100** dataset (mean ± standard deviation, %) under different image sizes and adversarial attack strengths ϵ.

Drop Ratio	Flower102	ϵ							
		0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08
0.0	Train	95.58±0.60	96.26±0.82	95.19±0.50	94.30±0.53	92.74±0.58	92.43±0.27	90.71±0.64	90.88±0.26
	Test	75.79±0.55	75.40±0.89	73.01±1.11	70.64±0.93	69.93±0.19	68.89±1.04	66.51±1.39	65.90±0.54
0.1	Train	95.91±0.66	94.88±0.23	94.66±0.24	94.50±0.28	93.30±0.40	92.24±0.79	91.50±0.58	90.67±0.85
	Test	77.25±0.48	74.62±0.71	73.71±0.49	71.11±0.76	70.85±0.57	69.36±0.42	67.74±0.83	65.39±0.75
0.2	Train	95.76±0.41	94.44±0.43	94.40±0.15	93.25±0.50	93.06±0.79	92.11±0.59	90.74±0.78	90.25±1.15
	Test	77.55±0.63	75.10±0.63	73.66±0.54	71.63±0.93	70.99±0.89	69.51±1.04	67.40±0.83	65.19±0.44
0.3	Train	95.27±0.11	94.69±0.41	93.49±0.25	92.26±0.99	92.75±0.37	90.95±0.76	89.30±0.95	90.61±0.89
	Test	77.41±0.68	74.90±0.51	72.38±1.56	70.28±1.78	70.42±1.68	68.87±1.06	66.58±0.77	66.42±0.77
0.4	Train	95.27±0.40	93.62±0.46	93.82±0.58	92.30±0.57	91.65±1.02	90.58±0.22	89.51±0.22	88.73±1.32
	Test	77.32±0.78	73.90±1.10	73.48±0.90	70.93±0.52	68.99±0.61	67.74±0.40	65.03±0.75	63.86±1.07
0.5	Train	94.44±0.45	93.83±0.32	91.06±1.16	91.58±1.73	91.32±0.42	89.04±1.31	87.68±1.73	87.26±1.89
	Test	75.65±1.50	73.57±1.35	70.83±1.15	69.29±1.59	69.91±1.01	66.79±0.79	64.82±1.03	64.49±1.03

Table 6: Adversarial training and testing accuracy (mean ± standard deviation, %) on **Flower102** dataset with DropKey regularization under different drop ratios and adversarial attack strengths ϵ.

Drop Ratio	Pets37	ϵ							
		0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08
0.0	Train	93.21±0.76	92.45±0.48	90.61±1.41	91.13±0.25	89.65±0.46	89.15±0.90	87.46±0.24	86.28±0.60
	Test	59.87±0.69	58.00±0.61	53.26±0.88	51.76±0.68	48.97±0.44	47.29±0.67	45.10±0.19	43.13±1.28
0.1	Train	93.16±0.34	91.28±0.45	91.63±0.28	90.87±0.35	88.97±0.47	87.65±0.45	86.39±0.70	84.82±0.28
	Test	60.00±0.68	55.90±1.00	53.96±0.64	50.84±0.64	49.79±0.16	47.47±0.80	45.54±1.36	39.73±0.76
0.2	Train	92.28±0.33	91.25±1.09	90.72±0.88	89.74±0.47	87.21±1.39	84.01±2.02	82.75±1.63	81.79±0.15
	Test	59.90±0.87	56.88±0.60	54.13±0.24	50.90±1.07	48.13±0.25	45.46±0.55	44.06±0.28	42.72±0.35
0.3	Train	91.58±1.24	90.64±0.36	89.72±0.20	86.25±0.61	85.15±0.30	83.77±1.13	81.39±1.18	80.85±0.18
	Test	59.96±1.14	55.39±0.20	52.84±0.42	50.08±0.61	46.97±1.47	45.88±0.19	43.79±0.70	43.73±0.73
0.4	Train	91.74±0.13	89.84±0.50	88.64±0.60	85.70±1.40	82.26±0.64	80.65±1.20	79.29±2.45	77.81±0.76
	Test	59.18±0.50	54.84±0.56	53.26±0.87	49.46±0.59	46.90±0.41	44.81±0.22	43.40±0.48	41.61±0.99
0.5	Train	88.75±0.76	89.28±0.16	67.45±12.02	73.97±0.28	81.28±0.60	76.68±1.73	66.00±0.11	66.55±0.21
	Test	58.15±0.27	56.77±0.20	36.76±9.41	38.20±9.12	46.76±0.59	46.09±1.75	33.75±5.99	32.13±5.26

Table 7: Adversarial training and testing accuracy (mean ± standard deviation, %) on **Pets-37** dataset with DropKey regularization under different drop ratios and adversarial attack strengths ϵ.

Drop Ratio	CIFAR10	ϵ							
		0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08
0.0	Train	94.83±0.15	94.06±0.18	92.92±0.18	90.86±0.19	90.11±0.34	89.99±0.18	87.87±0.13	86.41±0.16
	Test	82.14±0.17	80.18±0.37	77.87±0.12	75.60±0.12	74.33±0.34	73.03±0.11	71.67±0.11	70.18±0.30
0.1	Train	94.58±0.20	93.77±0.25	92.51±0.18	91.04±0.12	89.54±0.35	88.06±0.52	86.51±0.54	84.02±0.35
	Test	82.29±0.67	80.33±0.34	77.78±0.13	75.46±0.25	74.37±0.04	73.17±0.55	71.66±0.18	69.17±0.27
0.2	Train	94.41±0.20	93.33±0.39	91.43±0.27	90.31±0.70	88.67±0.53	86.92±0.17	85.50±0.27	83.52±0.21
	Test	82.59±0.44	79.99±0.31	77.81±0.18	75.77±0.49	74.26±0.36	72.68±0.31	71.00±0.05	69.88±0.17
0.3	Train	94.22±0.35	92.21±0.19	91.07±0.41	89.36±0.37	87.40±0.45	85.88±0.52	83.52±0.12	82.62±0.05
	Test	82.37±0.25	79.21±0.37	77.55±0.56	75.68±0.28	73.75±0.27	72.36±0.39	70.61±0.28	69.58±0.12
0.4	Train	93.39±0.15	91.97±0.20	90.46±0.29	88.28±0.32	85.71±0.13	83.40±0.03	82.82±0.67	80.12±0.28
	Test	82.33±0.09	79.83±0.19	77.97±0.49	75.44±0.26	73.15±0.23	71.21±0.06	70.93±0.09	67.56±0.33
0.5	Train	92.70±0.61	90.93±0.45	89.25±0.67	87.73±0.12	84.60±0.47	82.46±0.50	80.22±0.44	77.98±0.28
	Test	81.97±0.59	79.56±0.36	77.35±0.61	75.33±0.11	72.90±0.28	71.29±0.44	69.30±0.09	67.85±0.51

Table 8: Adversarial training and testing accuracy (mean ± standard deviation, %) on **CIFAR-10** dataset with DropKey regularization under different drop ratios and adversarial attack strengths ϵ.

Drop Ratio	Cifar100	ϵ							
		0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08
0.0	Train	92.71±0.43	91.30±0.31	89.56±0.45	88.47±0.18	86.22±0.51	85.20±0.38	82.90±0.73	82.11±0.40
	Test	56.97±0.40	52.54±0.19	49.95±0.23	47.22±0.61	45.01±0.24	43.28±0.26	41.85±0.10	40.53±0.57
0.1	Train	92.60±0.14	91.05±0.33	89.33±0.33	87.44±0.14	85.30±0.33	84.25±0.41	83.14±0.56	81.58±0.23
	Test	56.99±0.32	53.16±0.30	49.77±0.59	47.30±0.12	45.13±0.70	43.51±0.35	41.96±0.36	40.30±0.33
0.2	Train	92.38±0.26	90.44±0.42	88.73±0.43	86.55±0.47	84.58±0.52	82.61±0.29	81.83±0.42	80.21±0.08
	Test	56.91±0.25	52.90±0.15	49.78±0.30	47.49±0.62	44.70±0.29	43.21±0.26	41.56±0.59	40.45±0.41
0.3	Train	91.90±0.34	90.06±0.32	87.61±0.55	85.62±0.09	83.09±0.47	80.20±0.21	79.93±0.50	79.05±0.51
	Test	57.32±0.10	54.97±0.12	50.78±0.08	47.25±0.23	44.84±0.80	44.23±0.28	44.70±0.25	40.76±0.01
0.4	Train	91.50±0.19	89.17±0.45	87.00±0.33	83.79±0.44	81.56±0.08	80.64±0.10	79.27±0.36	77.73±0.70
	Test	56.91±0.24	52.99±0.22	49.88±0.12	47.02±0.19	44.79±0.30	43.08±0.40	41.86±0.04	41.24±0.21
0.5	Train	90.59±0.29	87.89±0.06	85.01±0.07	84.53±0.51	80.91±0.28	79.82±0.44	78.07±0.63	76.93±0.49
	Test	56.76±0.10	52.81±0.36	49.64±0.46	46.94±0.37	46.45±0.25	42.47±0.27	41.99±0.13	40.71±0.29

Table 9: Adversarial training and testing accuracy (mean ± standard deviation, %) on **CIFAR-100** dataset with DropKey regularization under different drop ratios and adversarial attack strengths ϵ.

References

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